Mihir Mistry – General Education Reflection

General education courses provide knowledge of topics past just technical skills learned in engineering courses. Although engineering focuses on math, science, and problem solving, engineers also need to know and understand topics such as economics and psychology while also learning about environmental and global factors. At Iowa State University, my general education courses have helped me recognize how engineering decisions affect society and sometimes need more than just a technical solution. Through the courses I have taken, which are Econ 101(Principles of Microeconomics), Psych 101(Introduction to Psychology), and ENVS 324(Energy and the Environment), I have learned more about the factors that impact engineering decisions. I also expect to take RUS 375(Russia Today), which I hope will expand my understanding of Russia and global relationships.

Econ 101 has helped me understand the importance of resources, costs, and tradeoffs when making decisions. Throughout the class, I learned that people have limited resources and they must make choices on how to use them. Every decision has a tradeoff, which means that choosing one option may mean giving something else up. This helped me think about how people decide when there is no perfect choice. This relates to engineering as engineers frequently need to choose between different designs/solutions, with each one having different tradeoffs. A design could be great, but not practical to use. The class itself may have been about economics, but it has helped me understand that engineering also involves finding a solution that involves costs, resources, and tradeoffs.

ENVS 324 has helped me understand the relationship between energy and its effects on the environment. I learned more about the different ways energy is produced and used, as well as how each type of energy affects the environment. This class helped me realize how important energy is and how choosing the right energy source can have lasting environmental impacts. I can relate this to engineering because engineers play a big role in developing energy systems that need to be both effective and sustainable. For example, climate change is a major problem that requires engineers to develop solutions to meet energy needs while helping reduce environmental impacts. What I learned in this class has helped me understand that the solutions engineers create should help society but should also consider the environmental impacts.

Psych 101 has helped me understand parts of human behavior and the factors that influence how people think, feel, and act. Throughout the class, I learned about the brain and how people's actions are influenced by both psychological and environmental factors, and how people may act differently. I can relate this to engineering because engineers create products and systems that people use all the time. Understanding people helps engineers create products that better suit people's needs. This class has taught me to think about how problem solving should consider the people that will use it.

RUS 375 is a class I am currently preparing to take that I expect to help me understand Russia's political, economic, and social impacts. I look forward to learning about how different factors

influence a country's development and its role in the world. I think this class will help me look at global issues from a different perspective. For example, international conflicts can disrupt the availability of resources needed for engineering projects. I can relate this to engineering because engineering projects can be influenced by international relationships, cultural differences, and the economy. By learning about Russia, I hope to understand more about another country while also learning about global issues and how things beyond engineering can influence engineering decisions.

Together, these courses show how general education helps solve engineering problems. Econ 101 helps create a foundation for looking at costs and trade-offs, ENVS 324 helps look at environmental consequences, PSYCH 101 makes us think about human behavior, and RUS 375 is expected to provide some global context. These classes have helped compliment my technical knowledge I've learned from my engineering work. General education has helped me realize that being a good engineer is not just finding a solution that works but finding one that is practical and beneficial to society.